
Claims development & reserving.

Estimating outstanding reserves using
Chain-Ladder and Random Forest approaches

GROUP D

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Data Science for Non-Life Insurance

THE QUESTION

How much will already-incurred claims ultimately cost?

RBNS

Reported but not yet settled

Claims known to the insurer but not yet fully settled. We observe the history, we need to project the tail

IBNR

Incurred but not reported

Claims unknown to the insurer, not yet reported. We have no history, we need to estimate their existence.

We start from the actuarial standard, expose its limitations, and complement it with an individual-claim Random Forest: comparing and combining both approaches.

The portfolio at a glance

33,977

(CLAIM, DEV-YR) RECORD

84,252

UNIQUE CLAIMS

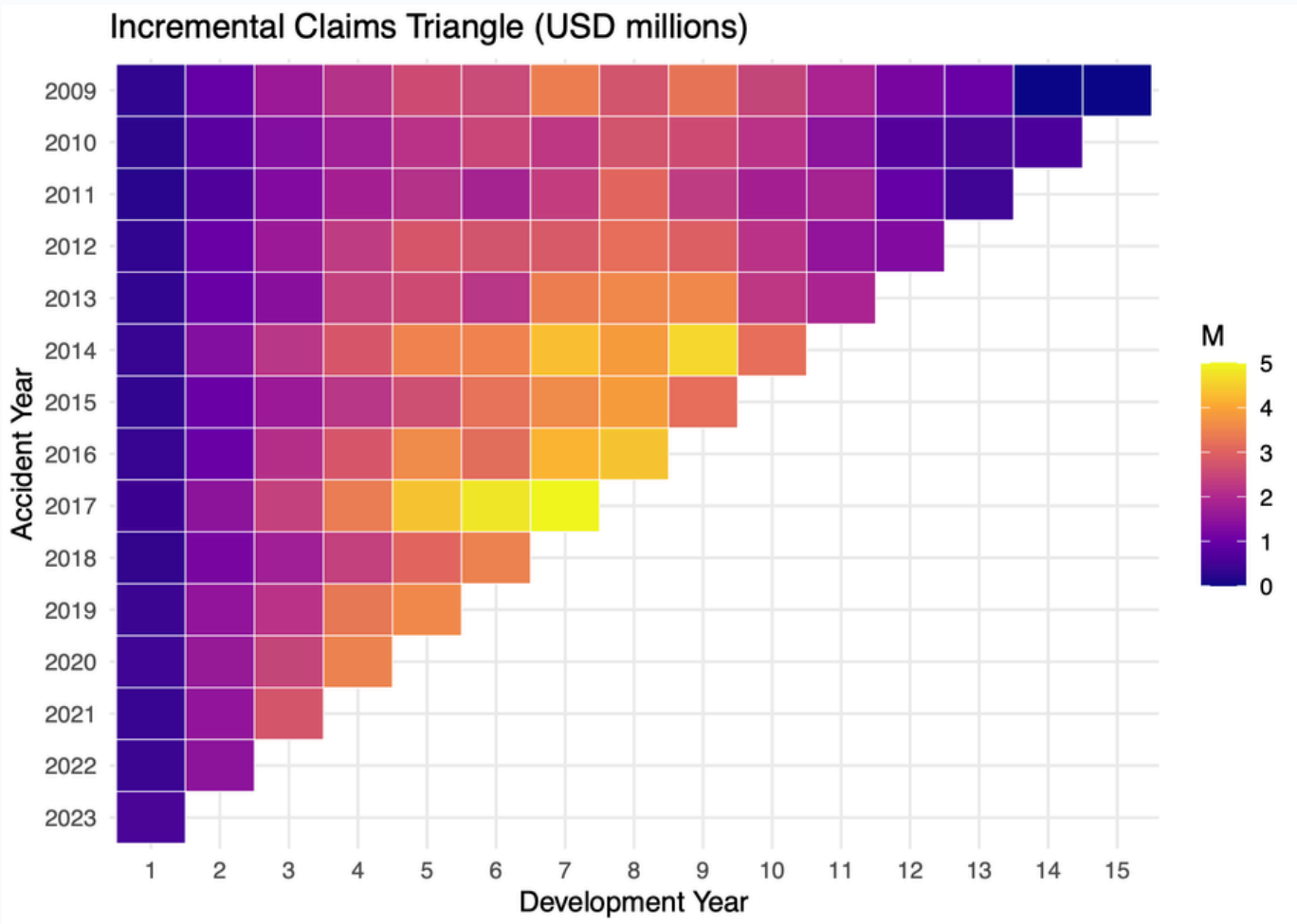
2009-2023

ACCIDENT YEARS

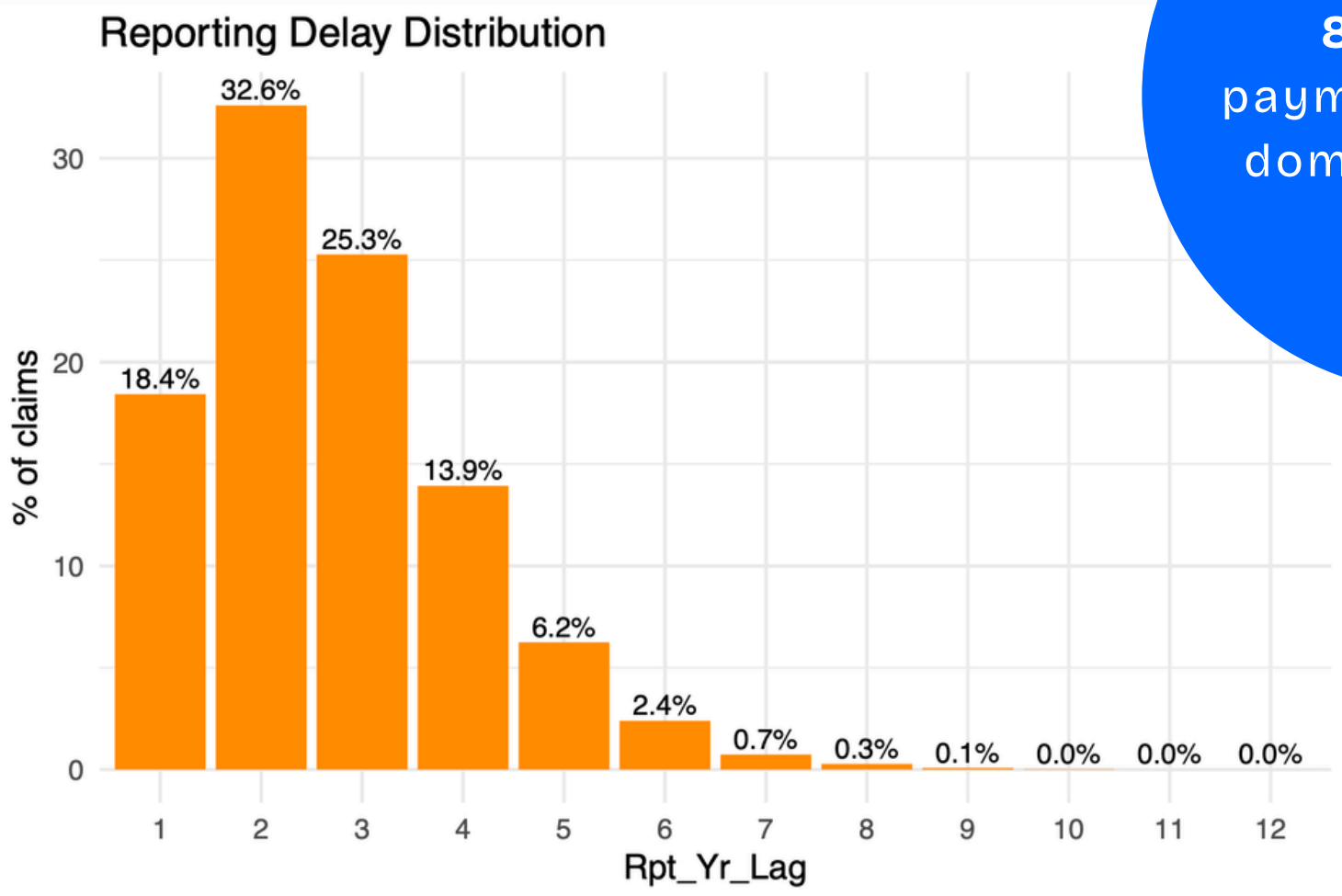
15

DEVELOPMENT
YEARS

RUN-OFF TRIANGLE



REPORTING DELAY



Rare but extreme:
80% zero
payments, top **1%**
dominates total
volume

Chain-Ladder

UNADJUSTED

USD 265 M

**INFLATION
ADJUSTED**

USD 261 M

+ super inflation

moderate (4%)

USD 315 M

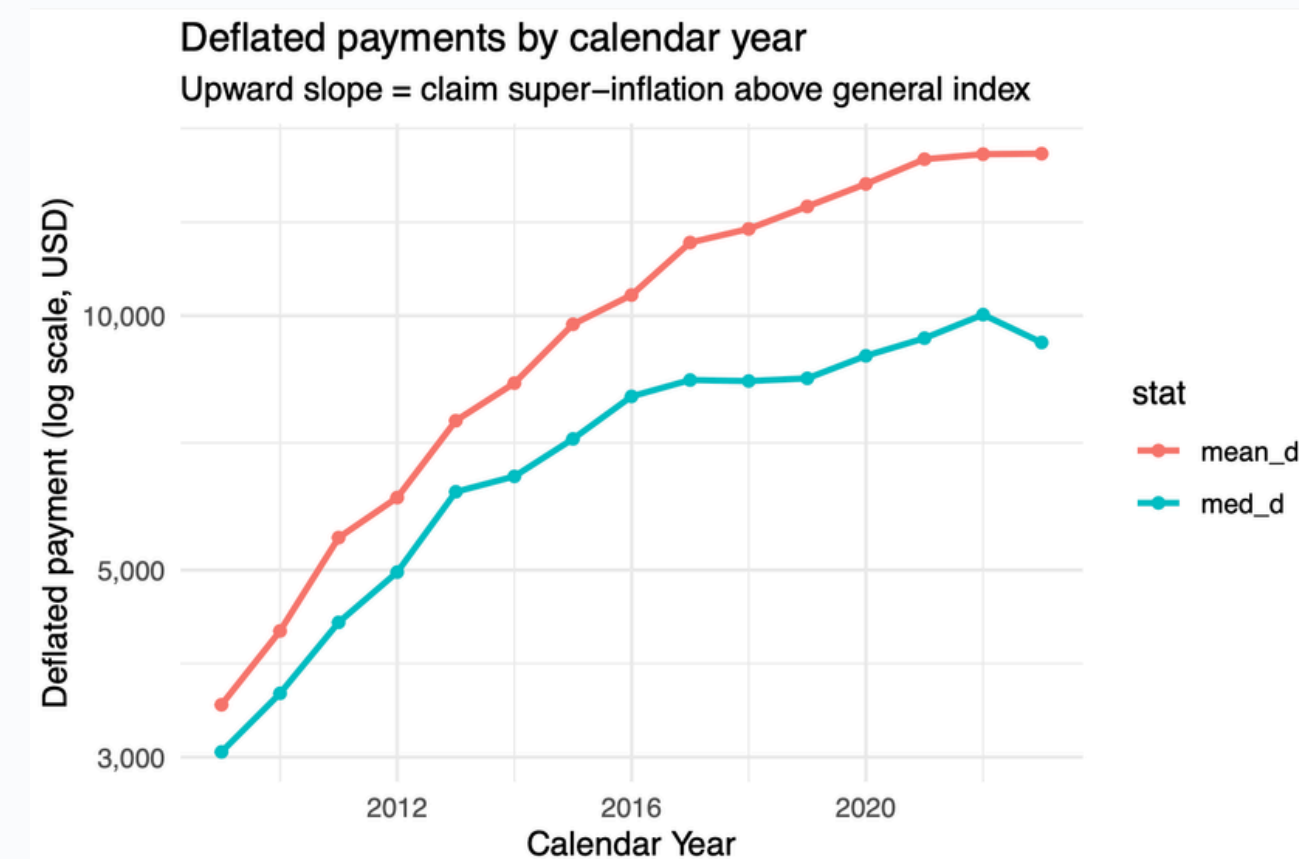
full (8%)

USD 382 M

Model assumptions



- (A1) independence between accident years
- (A2) stability of development factors over time
- (A3) variance proportional to the mean



Per-claim model: frequency × severity × closure

For each open claim and each future development year:

PART 1

Payment probability

Classifier with balanced weights
(≈ 80% zero-payment rate)

$\hat{p}_{\text{pay}}$

×

PART 2

Severity

Regression on log-deflated payment,
smearing-corrected on retransform

$\hat{\mu}_{\text{sev}}$

×

PART 3

Closure probability

Tracks effective lifetime;
avoids two-part overestimation

$\hat{p}_{\text{open}}$

$$\hat{S}_{i,k} = \hat{p}_{\text{open}} \cdot \hat{p}_{\text{pay}} \cdot \exp \left(\hat{\mu}_{\text{sev}} + \frac{\hat{\sigma}^2}{2} \right) \cdot I_{\text{infl}}$$

Initial RBNS-only estimate: **162 M** clearly too low

Why? Pure IBNR

THE IBNR GAP

Recent years are **still** arriving.

A forest trained on reported claims is blind to ones that haven't surfaced yet.

We estimate unreported counts from the empirical lag distribution (**mature AYs only**) and simulate them through the same **3-part model**.

$$N_{\text{IBNR}}(i) = N_{\text{obs}}(i) \cdot \frac{[1 - \Pr(\text{Lag} \leq t_i)]}{\Pr(\text{Lag} \leq t_i)}$$

RF • RBNS ONLY

162 Millions

+ 180 M ↓

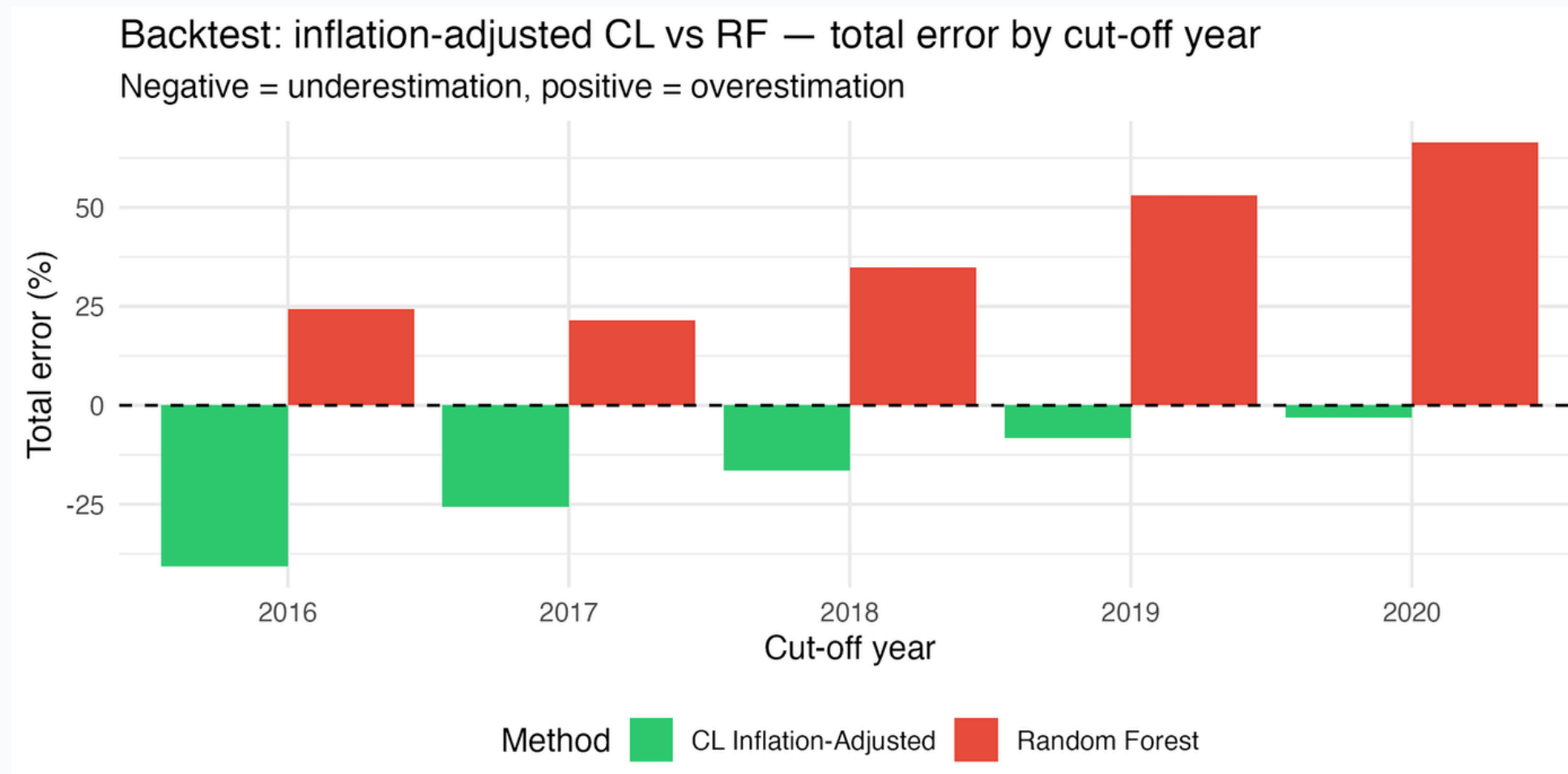
RF + IBNR

342 Millions

Now aligned with the chain-ladder envelope
The gap was missing claims, not model mis-fit

CL vs RF: **back-test**

Both models refitted at cutoffs 2016–2020, projected to actuals.



RF: systematic overestimation,
bias **grows** with more recent cutoffs
(+24% → +67%)

CL: systematic underestimation,
bias **shrinks** as horizon shortens
(-41% → -3%)

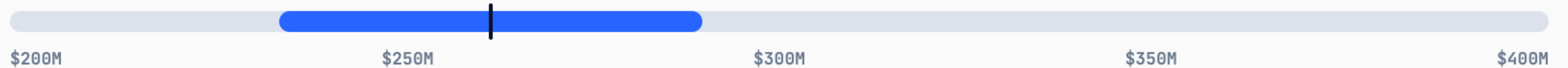
Opposite biases

How **wide** is the cone?

B=100 resamples. England-Verrall for CL
cluster bootstrap for RF.

CHAIN-LADDER • 95% CI

USD 235 – 290 M



RANDOM FOREST + IBNR • 95% CI

USD 265 – 345 M



Let dispersion decide.

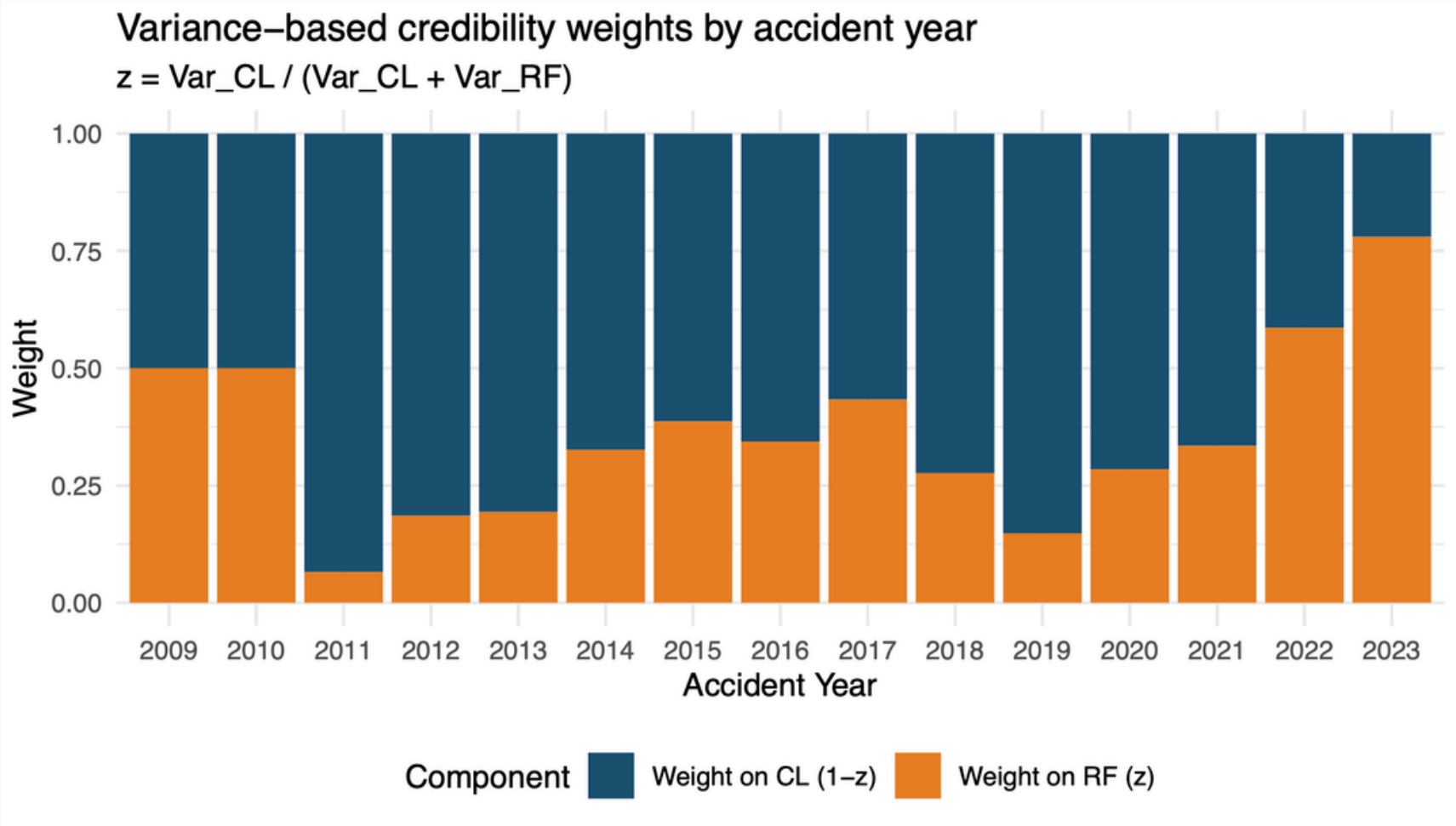
For each accident year, lean on whichever method is more stable across bootstrap replicates.

$$\hat{R}_i^{\text{blend}} = (1 - z_i) \hat{R}_i^{\text{CL,adj}} + z_i \hat{R}_i^{\text{RF+IBNR}}$$

$$z_i = \frac{\widehat{\text{Var}}(\hat{R}_i^{\text{CL}})}{\widehat{\text{Var}}(\hat{R}_i^{\text{CL}}) + \widehat{\text{Var}}(\hat{R}_i^{\text{RF}})}$$

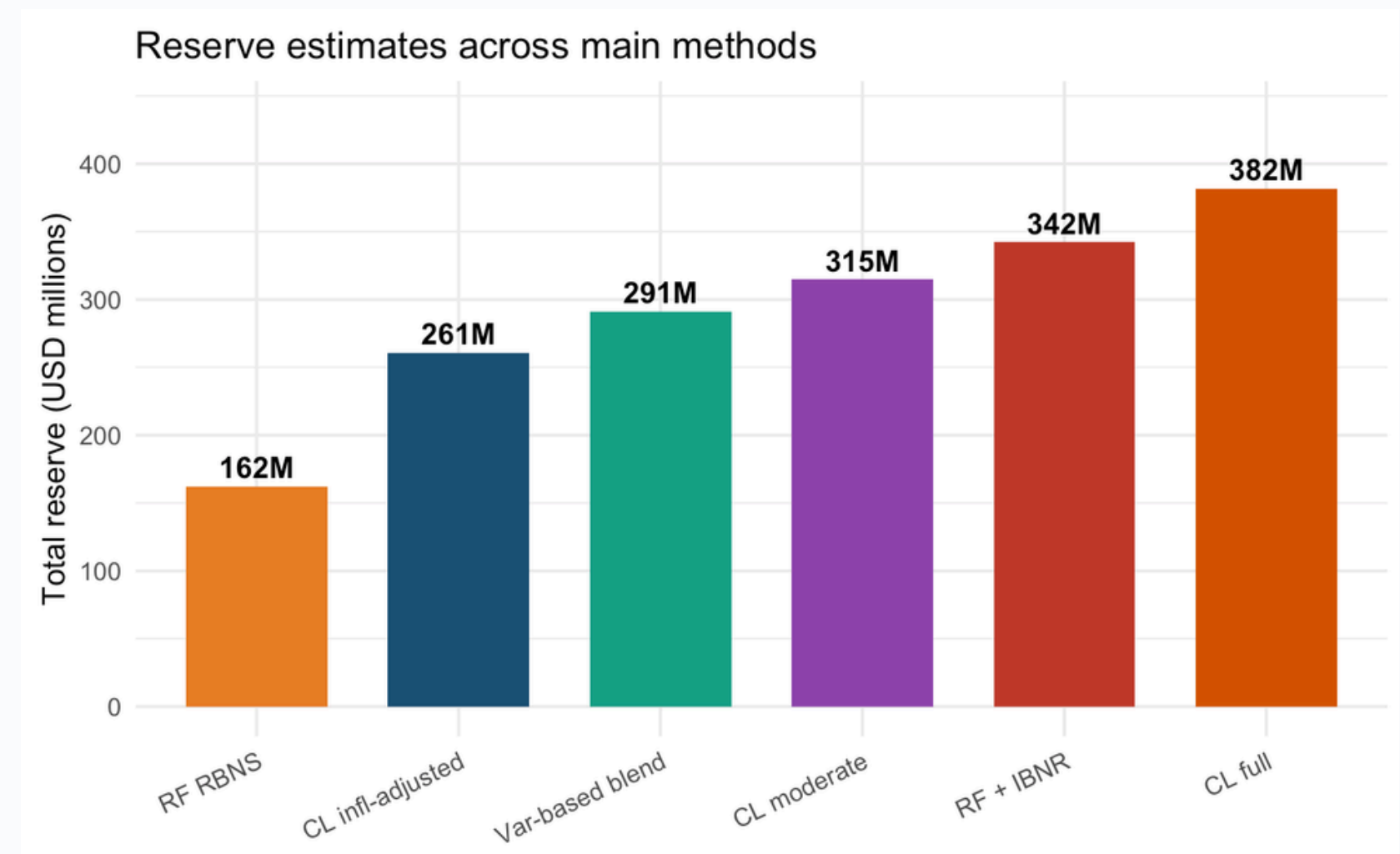
High CL variance → lean RF
High RF variance → lean CL

METHOD	RESERVE (USD)
CL inflation-adjusted	261 M
RF + IBNR	342 M
Variance-blend	291 M



Key takeaways.

- Chain-Ladder: **simple and transparent**, but blind to individual claim differences and highly sensitive to inflation assumptions.
- Random Forest: captures **claim-level heterogeneity**, but needs IBNR and censoring corrections to work.
- The variance-based blend combines both views, letting the more stable method lead.



central estimate of **280-340 M**

Thank you!
Any questions?